

## Advantages of cloud computing

- **Lower computer costs:** Since applications run in the cloud, one's desktop PC does not need the processing power or hard disk space demanded by traditional desktop software.
- **Improved performance:** Computers in a cloud computing system boot up and run faster because they have fewer programs and processes loaded into the memory.
- **Reduced software cost:** Instead of purchasing expensive software applications, you can get most of what you need for almost free.
- **Instant software updates:** When the application is Web based, updates happen automatically. That is, when you access a Web based application, you get the latest version.
- **Improved document format compatibility:** You do not have to worry about the documents you create on your machine being compatible with other users' applications or operating systems.
- **Unlimited storage capacity:** Cloud computing offers virtually limitless storage.
- **Increased data reliability:** If your personal computer crashes, all your data is safe out there in the cloud, still accessible.
- **Easier group collaboration:** Multiple users can collaborate easily on documents and projects.
- **Device independence:** You are no longer tethered to a single computer or network.

In spite of all the advantages of the cloud, stored data might not be secure. Since all your data is stored on the cloud, the important question is: How secure is the cloud? Can unauthorised users gain access to your confidential data?

When an organisation elects to store data or host applications on the public cloud, it loses the ability to have physical access to the servers hosting its information. As a result, sensitive and confidential data is at risk from outsider and insider attacks.

## Measures to be taken to improve the security of the cloud

- **Cloud service providers must ensure proper data isolation**  
In order to conserve resources, cloud service providers often store more than one client's data on the same server. As a result, there is a chance that one user's private data can be viewed by other users (possibly even competitors). To handle such sensitive situations, cloud service providers should ensure proper data isolation and logical storage segregation.
- **Encryption of data**

Enterprises must select a cloud storage provider that supports encryption of data in-flight and data at-rest. Amazon Web Services (AWS) moves data from Simple Storage Service (S3) across an SSL connection and protects that data using AES-256 encryption. Also, enterprises can choose available third-party encryption tools such as Viivo, Sookasa or Cloudfogger.

- **Data centres must be frequently monitored**

According to a recent report, insider attacks are the third biggest threat in cloud computing. Therefore, cloud service

providers must ensure that thorough background checks are conducted for employees who have physical access to the servers in the data centre. Additionally, data centres must be frequently monitored for suspicious activity.

- **Use of virtualisation should be reduced**

Virtualisation alters the relationship between the OS and underlying hardware - be it computing, storage, or even networking. The extensive use of virtualisation in implementing cloud infrastructure brings along unique security concerns for customers of a public cloud service. It is said that even a breach in the administrator's workstation, by the management software of the virtualisation software, can cause the whole data centre to go down or be reconfigured to an attacker's liking.

## Open source cloud computing technologies

A few open source cloud computing technologies are briefly discussed in this section.

### OpenStack

OpenStack comprises a set of software tools for building and managing cloud computing platforms for public and private clouds. It is one of the most important open source technologies for enterprises and developers. OpenStack is considered as Infrastructure as a Service (IaaS). It provides infrastructure, making it easy for users to quickly add new instances, upon which other cloud components can run. The infrastructure then runs a platform upon which a developer can create software applications that are delivered to the end users. OpenStack serves a continuously increasing number of IT environments as a foundation for public, private and managed infrastructure. Organisations in particular have used OpenStack to build their own private clouds. For instance, Deutsche Telekom (Business Marketplace) uses OpenStack to build its cloud platforms.

If you want to use OpenStack, first use tryStack, which will let you test your applications in a sandbox environment. This will enable you to understand how OpenStack works and whether it is the right solution for you.

### Cloud Foundry

In the growing Platform-as-a-Service (PaaS) market, Cloud Foundry takes a leading position. The project was initialised by Pivotal, a spin-off by EMC/VMware. Cloud Foundry is primarily written in Ruby and Go. Applications deployed to Cloud Foundry access external resources via services. In a PaaS environment, all external dependencies such as databases, messaging systems and file systems are services. When an application is pushed to Cloud Foundry, the services it should use can also be specified. Depending on the application language, auto-configuration of services is possible; for example, a Java application requiring a MySQL database picks up the MySQL service on Cloud Foundry if it is the only one defined in the current space.